

PROJECT MANUAL – IQA CHATBOT SYSTEM

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PROJECT OVERVIEW

The IQA Chatbot System is built to handle uploaded PDF documents by automatically parsing, chunking, and converting them into embeddings. These processed data chunks are then served via a chatbot interface powered by the Mistral 7B model through Ollama. The architecture is lightweight, modular, and easily extendable — enabling straightforward upgrades to both the backend logic and the language model with minimal changes.

DATABASE STRUCTURE (MONGODB)

Database Name: iqa

The system uses the following collections in the MongoDB database:

Collection Name	Description
activities	Tracks user activity and system-level events.
apis	Stores API endpoints and configurations including the Mistral Ollama setup.
chats	Saves user-chatbot interactions for history or analytics.
logs	General system logs.
logs_admins	Specific logs related to admin operations.
logs_sysadmins	Logs related to sysadmin functions and errors.
logs_users	User interaction logs.
logs_visitors	Tracks Non-Admin Users who are solely entitled to use the chatbot only.
portable	Contains uploaded PDF files and their chunked/preprocessed data.
settings	Stores platform-wide or model-specific configuration.
trash	Temporary or deleted data.
trashes	Backup or log of deleted entries for recovery/audit.
updates	Tracks system updates or model changes.
users	User account data and roles.

PDF UPLOAD AND PROCESSING

Uploaded PDFs are handled in the portable collection. Each file is automatically:

- Stored in the database
- Parsed and chunked
- Preprocessed into embeddings usable by the chatbot

LLM INTEGRATION – OLLAMA + MISTRAL 7B

The system uses Mistral 7B, integrated via Ollama.

Setup Steps

1. Install Ollama from: <https://ollama.com>

2. Download the model:

```
ollama pull mistral
```

3. Verify Ollama is active:

```
ollama ps
```

4. Serve the model:

```
ollama run mistral
ollama list
```

Model Switching

The apis collection contains the model endpoint (<http://localhost:11434/api/generate>) and configuration JSON. Switching models only requires updating these fields—no backend changes required.

MODEL SPECIFICATIONS & SYSTEM REQUIREMENTS

Model	Parameters	Quantization	VRAM (Min/Recommend)	Use Case
Mistral 7B	7B	Q4	6 GB / 8 GB	Fast and accurate for general usage
all-minilm:latest	1B	N/A	512 MB / 1 GB	Lightweight; good for sentence embeddings
nomic-embed-text:latest	1B+	N/A	1 GB / 2 GB	Versatile embedding model; useful for large document search

DEPLOYMENT ON UTM SERVER

Important: This project is complete. The following steps cover only placing the system on a production server.

1. Access a Linux server (Ubuntu 20.04 or 22.04 preferred)

2. Clone the repository:

```
git clone https://github.com/Tanshiba12/IQA_tail.git
```

```
cd IQA_tail
```

3. Install Ollama:

```
curl -fsSL https://ollama.com/install.sh | sh
ollama serve &
```

4. Pull the model:

```
ollama pull mistral:7b-instruct-q4
```

5. Create .env with:

```
PORT=5000
OLLAMA_API=http://localhost:11434
MONGO_URI=<MongoDB URI from handover>
MODEL=mistral
```

6. Install dependencies:

- Python: pip install -r requirements.txt
- Node: npm install

7. Run backend:

- Python: python server.py
- Node: node index.js

8. Build frontend (if applicable):

```
cd frontend
npm install
npm run build
```

9. Host with serve, nginx, or another static file server.

10. Optional: Setup PM2 or systemd for persistence.

SYSTEM REQUIREMENTS — CPU-BASED

Params (B)	Quantization	Bits	Disk Space	RAM Re-quired	CPU Cores (Threads)	Notes
2B–3B	Q4 / Q5	4–5	1.5–2.5 GB	8–12 GB	4–6 cores (8–12 threads)	Fastest CPU-only performance
7B	Q4 / Q5	4–5	4–6 GB	12–16 GB	8–12 cores (16–24 threads)	Usable; some delay (2–4s replies)
13B	Q4	4	8–10 GB	24–32 GB	12–16 cores (24–32 threads)	Not ideal without GPU

30B+	Q4	4	15–30 GB	48–64 GB+	24+ cores	Not recommended for CPU-only environments
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SYSTEM REQUIREMENTS — GPU-BASED (HYBRID DEPLOYMENT)

Params (B)	Quantization	Bits	Disk Space	RAM	GPU VRAM (Min/Ideal)	Notes
2B–3B	Q4 / Q5	4–5	1.5–2.5 GB	8–12 GB	4 GB / 6 GB	Fast replies for multiple users
7B	Q4	4	4–5 GB	12–16 GB	8 GB / 12–16 GB	Great speed with RTX 3060 or better
7B	Q6_K	6	6–7 GB	16–24 GB	12–24 GB	Higher accuracy, needs strong GPU
13B	Q4	4	9–10 GB	24–32 GB	16–24 GB	Recommended for research or full rollout
30B	Q4	4	15–20 GB	48–64 GB	24–40 GB	Data center-grade GPU required
65B	Q4	4	30–45 GB	80 GB+	40–80 GB	For research only — cluster or multi-GPU only

FINAL NOTES FOR DEPLOYMENT

- MongoDB is already set up; do not recreate the schema.
- The project is handover-ready; no additional setup is required.
- The model can be upgraded by changing the configuration in the `apis` collection.
- Logs and analytics are saved in the database; the system is self-contained.
- Request GPU allocation if higher speed or concurrency is needed.